

# Topology PhD Exam, May 2020

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Work the following problems and show all work. Support all statements to the best of your ability.

- Let  $X$  be a nonempty set. A nonempty collection  $\Phi$  of subsets  $U \subseteq X \times X$  is a uniformity if it satisfies the following axioms:
  - If  $U \in \Phi$ , then  $\Delta \subseteq U$ , where  $\Delta$  is the diagonal  $\{(x, x) : x \in X\}$ .
  - If  $U \in \Phi$  and  $U \subseteq V \subseteq X \times X$ , then  $V \in \Phi$ .
  - If  $U, V \in \Phi$ , then  $U \cap V \in \Phi$ .
  - If  $U \in \Phi$ , then there is a  $V \in \Phi$  with  $V \circ V \subseteq U$ , where  $A \circ B = \{(x, z) \mid \exists y \in X \text{ such that } (x, y) \in A \text{ and } (y, z) \in B\}$ .
  - If  $U \in \Phi$ , then  $U^{-1} \in \Phi$ , where  $U^{-1} = \{(y, x) \mid (x, y) \in U\}$ .

The elements of  $\Phi$  are called entourages. If  $x \in X$  and  $U \in \Phi$ , set  $U[x] = \{y \mid (x, y) \in U\}$ . A uniformity induces a topology on  $X$ : a set  $O$  is open if for every  $x \in O$  there is an entourage  $V$  such that  $V[x] \subseteq O$ . A topological space is uniformizable if there is a uniformity compatible with the topology (that is, the topology generated by the uniformity is the topology on  $X$ ).

Prove that a metric space  $X$  is uniformizable.

- Let  $T$  be the 2-dimensional torus. For each of the following, give an example or prove such an object does not exist.
  - A map  $f : T \rightarrow S^2$  of degree 1.
  - A map  $g : S^2 \rightarrow T$  of degree 1.
- The torus  $T$  bounds a compact region  $R$ . Two copies of  $R$  are glued together by identifying their boundaries via the map  $f : T \rightarrow T$  given by  $f(\theta, \varphi) = (\theta + \varphi, \theta + 2\varphi)$  (here  $T = S^1 \times S^1$ , where points on  $S^1$  are identified with an angle  $\theta \in [0, 2\pi)$ ). The result is a compact 3-manifold  $M$ .
  - Compute the map  $f_* : H_1(T) \rightarrow H_1(T)$ .
  - Use the Mayer-Vietoris sequence to compute the homology of  $M$ .
- Let  $X \subset \mathbb{R}^3$  be the union of  $n$  lines through the origin. Compute  $\pi_1(\mathbb{R}^3 - X)$ .
- Compute  $H^\bullet(M; \mathbb{Z})$ , where  $M$  is the space with homology groups

$$H_k(M; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, 2, 4, \\ \mathbb{Z}_3, & k = 1, \\ \mathbb{Z}_2, & k = 3, \\ 0, & \text{otherwise.} \end{cases}$$

Answer the following with complete definitions, statements, or short proofs.

- Prove that for a finite CW-complex  $X$ ,  $H^1(X; \mathbb{Z})$  is free.
- Compute  $\chi(\mathbb{CP}^2 \times \mathbb{RP}^4 \times S^2)$ .

8. Give an example of a space that is connected but not path-connected.
9. State the Urysohn Metrization Theorem.
10. Prove that if  $m \neq n$ , then  $\mathbb{R}^m$  is not homeomorphic to  $\mathbb{R}^n$ .
11. Let  $X$  be a space and let  $X^*$  be its one-point compactification. Prove that if  $X$  is locally compact and Hausdorff, then  $X^*$  is Hausdorff.
12. State the Lefschetz Fixed Point Theorem.
13. Describe all the connected covering spaces  $E \rightarrow S^1 \times S^1$ .
14. Does the following exact sequence of abelian groups necessarily split? Prove or give a counterexample.

$$0 \rightarrow A \rightarrow B \rightarrow \mathbb{Z} \rightarrow 0$$

15. Compute the integral homology of the space  $\mathbb{RP}^2 \times \mathbb{CP}^3$ .